EXHIBIT E

DEPARTMENT OF HEALTH & HUMAN SERVICES

Office of the Secretary

Office of the Assistant Secretary for Health
Washington, D.C. 20201

OCT 17 2017

The Honorable Robert W. Patterson
Acting Administrator
Drug Enforcement Administration
U.S. Department of Justice
8701 Morrissette Drive
Springfield, VA 22152

Dear Mr. Patterson:

Pursuant to the Controlled Substances Act (CSA), 21 U.S.C. § 811 (b), (c), and (f), the Department of Health and Human Services (HHS) is recommending that the substances mitragynine and 7-OH-mitragynine be permanently controlled in Schedule I of the CSA. Mitragynine and 7-OH-mitragynine are mu opioid receptor agonists that represent two of the principle psychoactive constituents of the plant *Mitragyna speciosa (M. speciosa)*, commonly referred to as ***kratom***. In assessing the relative abuse potential of mitragynine and 7-OH-mitragynine, FDA reviewed and evaluated all of the available data on the abuse potential of mitragynine and 7-OH-mitragynine. In order to conduct a full evaluation of mitragynine and 7-OH-miyragynine, it was necessary to review information on the use of *M. speciosa*, the only natural source of mitragynine and 7-OH-mitragynine. Identification of *M. speciosa* cannot be accomplished solely through chemical analysis of mitragynine and 7-OH-mitragynine content alone. This is because adulteration of botanical material from any plant with mitragynine and 7-OH-mitragynine would make its chemical analysis indistinguishable from that of mitragynine and 7-OH-mitragynine from *M. speciosa*. Thus, our scheduling recommendation does not include a recommendation for scheduling *M. speciosa* due to the difficulty in identifying the plant.

Mitragynine and 7-OH-mitragynine are mu opioid agonists that elicit pharmacological responses that are similar to those produced by other mu opioid agonists, including morphine (Schedule II). In animals, mitragynine and 7-OH-mitragynine produce full generalization to morphine in drug discrimination and produce conditioned place preference, indicating these substances are rewarding. In humans, mitragynine and 7-OH-mitragynine (typically as constituents of *M. speciosa)* produce classic opioid effects such as euphoria, sedation, and analgesia. These substances also produce an opioid withdrawal syndrome upon drug discontinuation. Thus, mitragynine and 7-OPH-mitragynine have a high potential for abuse.

The Food and Drug Administration (FDA) and the National Institute on Drug Abuse have also considered the abuse potential of mitragynine and 7-OH-mitragynine. After reviewing the available information, the agencies conclude that mitragynine and 7-OH-mitragynine should be permanently controlled in Schedule I of the CSA. Enclosed is a document prepared by FDA's Controlled Substance Staff that forms the basis for the recommendation.

BASIS FOR THE RECOMMENDATION TO CONTROL MITRAGYNINE AND 7-HYDROXYMITRAGYNINE IN SCHEDULE I OF THE CONTROLLED SUBSTANCES ACT

On August 21, 2014, the Drug Enforcement Administration (DEA) requested that the Department of Health and Human Services (HHS) conduct a medical and scientific evaluation of mitragynine and 7-hydroxymitragynine (7-OH-mitragynine) to determine whether these substances should be recommended for control under the Controlled Substances Act (CSA). Mitragynine and 7-OH-mitragynine are mu opioid receptor agonists that represent two of the principle psychoactive constituents of the plant *Mitragyna speciosa (M. speciosa)*, commonly referred to as ***kratom***. DEA also requested an abuse potential evaluation of *M. speciosa*.

Subsequently, on August 31, 2016, DEA published a Notice of Intent (NOI) in the *Federal Register* "to temporarily schedule the opioids mitragynine and 7-OH-mitragynine into Schedule I, pursuant to the temporary scheduling provisions of the CSA." DEA stated that this action was taken "to avoid an imminent hazard to the public safety" (81 FR 59929). On October 13, 2016, DEA withdrew this NOI, stating that they had "received numerous comments from members of the public challenging the scheduling action and requesting that the agency consider those comments and accompanying information before taking further action." DEA also stated they had requested that "FDA expedite its scientific and medical evaluation and scheduling recommendation for these substances, which DEA previously requested" (81 FR 70652).

In assessing the relative abuse potential of mitragynine and 7-OH-mitragynine, FDA reviewed and evaluated all of the available data on the abuse potential of mitragynine and 7-OH-mitragynine. In order to conduct a full evaluation of mitragynine and 7-OH-mitragynine, it was necessary to review information on the use of *M. speciosa*, the only natural source of mitragynine and 7-OH-mitragynine. As detailed below, however, this scheduling recommendation does not include a recommendation for scheduling *M. speciosa*, due to the difficulty in accurately identifying the plant. Identification of *M. speciosa* cannot be accomplished solely through chemical analysis of mitragynine and 7-OH-mitragynine content alone. This is because adulteration of botanical material from any plant with mitragynine and 7-OH-mitragynine would make its chemical analysis indistinguishable from that of mitragynine and 7-OH-mitragynine from *M. speciosa*. Thus, obtaining unequivocal identification of *M. speciosa* can only be accomplished through the combined use of chemical analysis of mitragynine and 7-OH-mitragynine (through high-performance liquid chromatography-mass spectrometry) and molecular analysis of the plant substance (through DNA or RNA sequence analysis (Sukrong et al., 2007)). This type of confirmatory data related to the identification of *M. speciosa* alone is typically unavailable.

Thus, this document does not include a scientific and medical evaluation of botanical *M speciosa* and restricts the evaluation to two of its major psychoactive chemical components, mitragynine and 7-OH-mitragynine.

Pursuant to 21 U.S.C. 811(b) of the CSA, the Secretary of HHS is required to consider in a scientific and medical evaluation eight factors determinative of control under the CSA.

DNA or RNA sequence analysis, Sukrong et al, 2007). This type of confirmatory data related to the identification of *M. speciosa* alone is typically unavailable. Thus, this document does not include a scientific and medical evaluation of botanical *M. speciosa* and restricts the evaluation to two of its major psychoactive chemical components, mitragynine and 7-OH-mitragynine.

This review evaluates data from the medical and scientific literature, from federal government reports, and databases on the subjective responses and adverse events that result from use of mitragynine and 7-OH-mitragynine.

1. The Drug's Actual or Relative Potential For Abuse

The first factor the Secretary must consider is the actual or relative potential for abuse of mitragynine and 7-OH-mitragynine. The term "abuse" is not defined in the CSA. However, the legislative history of the CSA suggests the following points in determining whether a particular drug or substance has a potential for abuse[1]:

a. Individuals are taking the substance in amounts sufficient to create a hazard to their health or to the safety of other individuals or to the community.

b. There is significant diversion of the drug or substance from legitimate drug channels.

c. Individuals are taking the substance on their own initiative rather than on the basis of medical advice from a practitioner licensed by law to administer such substance.

d. The substance is so related in its action to a substance already listed as having a potential for abuse to make it likely that it will have the same potential for abuse as such substance, thus making it reasonable to assume that there may be significant diversions from legitimate channels, significant use contrary to or without medical advice, or that it has a substantial capability of creating hazards to the health of the user or to the safety of the community.

When assessing the abuse potential of a substance, the Secretary considers multiple factors, data sources, and analyses. These factors include the prevalence, frequency, and manner of use in the general public and specific subpopulations, the amount of material that is available for illicit use, as well as evidence relevant to populations that may be at particular risk.

Animal, human, and epidemiological data are all used in determining a substance's abuse potential. Scientifically, a comprehensive evaluation of the relative abuse potential of a substance generally includes consideration of the drug's receptor activity and neurochemical effects, discriminative stimulus effects, rewarding effects, pharmacokinetics, subjective

[1] Comprehensive Drug Abuse Prevention and Control Act of 1970, H.R. Rep. No. 91-1444, 91st Cong., Sess. 1 (1970) reprinted in U.S.C.C.A.N. 4566, 4603

and 7-OH-mitragynine have substantial capability to be a hazard to the health of the user and to the safety of the community.

Summary of Potential for Abuse of Mitragynine and 7-OH-mitragynine:

The following conclusions indicate that mitragynine and 7-OH-mitragynine have a potential for abuse:

a) There is evidence that individuals are taking mitragynine and 7-OH-mitragynine in amounts sufficient to create a hazard to their health or to the safety of other individuals or to the community.

b) Mitragynine and 7-OH-mitragynine are not available from legitimate drug channels other than for use in scientific research, and there is no evidence of significant diversion from such channels.

c) There is significant evidence from epidemiological databases and the scientific literature that individuals are using mitragynine and 7-OH-mitragynine for abuse purposes.

d) The pharmacological effects of mitragynine and 7-OH-mitragynine are similar to those of morphine, a Schedule II drug with high potential for abuse. This supports the conclusion that mitragynine and 7-OH-mitragynine present a hazard to the health and safety of individuals and the community.

2. SCIENTIFIC EVIDENCE OF THE DRUG'S PHARMACOLOGICAL EFFECTS, IF KNOWN

The second factor the Secretary must consider is scientific evidence of the pharmacological effects of mitragynine and 7-OH-mitragynine. The scientific evidence for these compounds include their receptor activity and functional effects, their effects in animal behavioral studies, their subjective effects in humans, and serious adverse events resulting from their use.

Central Nervous System Effects

Receptor Activity and Functional Effects

Mitragynine and 7-OH-mitragynine have been shown in binding studies to have high affinity for human mu opioid receptors (Ki = 233 nM and 47 nM, respectively) (Kruegel et al., 2016). Functional studies with these compounds showed that they are acting as partial agonists in human tissue, with maximal effects in G-protein activation of 34 percent for mitragynine and 47 percent for 7-OH-mitragynine (Kruegel et al., 2016). Mu opioid partial agonists are able to produce significant mu opioid agonist activity, as long as the dose is not increased beyond a certain level. At higher doses, a mu opioid partial agonist may act as an antagonist. If an individual has been using mu opioid agonist drugs repeatedly and has developed physical

agonist, morphine, was tested in the CPP test, it produced a similar preference in animals for the drug-associated side of the cage. These data suggest that mitragynine and 7-OH-mitragynine produce rewarding properties that are similar to that of a Schedule II mu opioid agonist.

Activity at Non-Opioid Receptors

Activation of 5HT2 receptors by 5-methoxy-dimethyltryptamine (a Schedule I hallucinogen) produces head twitches in mice. This behavior is known to be blocked by 5HT2 antagonists and by alpha-2 adrenergic antagonists. When mice were pre-treated with mitragynine, there was a dose-dependent reduction in 5-MeO-DMT-induced head twitches (Matsumoto et al., 1997). This suggests that mitragynine has antagonistic activity at 5HT2 or alpha-2 adrenergic receptors.

Development of Tolerance

Tolerance is a physiological adaptation in which repeated exposure to a specific dose of a drug will result in a lessening of the effects of that drug at the same dose (Cahill et al., 2016). The development of tolerance is a well-known phenomenon following chronic administration of mu opioid agonists such as morphine (Schedule II). Individuals who develop tolerance after long-term use of mu opioid agonists will often increase the dose of the drug in order to still obtain the desired effects, regardless of whether those effects are therapeutic in nature or include euphoria for abuse purposes. Since tolerance can differentially develop to the various effects of a mu opioid agonist, it is possible that an individual may be tolerant to the therapeutic or abuse responses produced by an opioid, but may not yet be tolerant to certain adverse events. This can lead to a situation where a higher dose of a mu opioid agonist taken to overcome tolerance may produce unexpected serious adverse events such as respiratory depression and death (Cahill et al., 2016).

Individuals who chronically use *M. speciosa* typically develop tolerance, leading to an increase their intake of the drug over time (Suwanlert, 1975). When mitragynine was repeatedly administered to mice, tolerance to the antinociceptive effects developed over the course of 5 days (Fakurazi et al., 2013). Similarly, when 7-OH-mitragynine was given to mice repeatedly for 5 days, there was a significant reduction in the antinociceptive response (Matsumoto et al., 2005). In each of these studies, the time course and degree of tolerance produced by mitragynine and 7-OH-mitragynine was similar to the development of tolerance produced by the Schedule II mu opioid agonist, morphine.

Cross-tolerance was also observed with 7-OH-mitragynine and morphine (Matsumoto et al., 2005). Thus, once tolerance developed to one drug (such that a larger dose was necessary to produce analgesia), then a larger dose of the other drug was also needed in that animal to produce analgesia. This phenomenon occurs when drugs have overlapping mechanisms of action. Thus, these results suggest that 7-OH-mitragynine has mu opioid agonist effects similar to that of morphine.

mitrasulgynine, 3,4,5,6-tetradehydromitragynine, corynantheidaline, corynantheidine, isocorynoxeine, 9-methoxymitralactonine, and mitralactonine.

Mitragynine is known chemically as (αE,2S,3S,12bS)-3-ethyl-1,2,3,4,6,7,12,12b-octahydro-8-methoxy-α-(methoxymethylene)-indolo[2,3-a]quinolizine-2-acetic acid methyl ester (CAS number 4098-40-2). It has a molecular formula of $C_{23}H_{30}N_2O_4$; a molecular weight of 398.5; and is soluble in acetone, acetic acid, alcohols, chloroform, and diethyl ether. It is found as white, amorphous crystals with a melting point of 102–106 °C.

7-OH-mitragynine is known chemically as (αE,2S,3S,7aS,12bS)-3-ethyl-1,2,3,4,6,7,7a,12b-octahydro-7a-hydroxy-8-methoxy-α-(methoxymethylene)-indolo[2,3-a]quinolizine-2-acetic acid methyl ester (CAS number 174418-82-7). It has a molecular formula of $C_{23}H_{30}N_2O_5$, a molecular weight of 414.5. The solubility, appearance, and melting point of this substance were not characterized.

The first total synthesis of mitragynine was performed by Takayama et al. (1995). Subsequently, a complete synthesis of mitragynine was described using 4-methoxy-tryptophan (Ma et al., 2007; Ma et al., 2009), an uncommon starting material. Total synthesis of mitragynine involves up to 23 steps, is time-consuming, and is not economical because it produces a low yield (~ 3–13%) (Kerschgens et al., 2012). Thus, total synthesis of mitragynine is unlikely to occur in order to support an illicit market. Therefore, use of botanical *M. speciosa*, or isolation of mitragynine and 7-OH-mitragynine from *M. speciosa*, is a more accessible source of the drug for users.

Levels of Mitragynine and 7-OH-Mitragynine in M. speciosa

The content of mitragynine found naturally in *M. speciosa* is variable. The total alkaloid content in *M. speciosa* leaves usually ranges from 0.5 to 1.5 percent (Hassan et al., 2013). Plants from Thailand were found to have a mitragynine content of 66 percent of the total alkaloid contents (or approximately 3-9 mg/g in the raw leaves), while leaves from trees in Malaysia were found to have a mitragynine content of 12 percent of the total alkaloid contents (Takayama et al., 1998; Chittrakarn et al., 2008; Ponglux et al., 1994; Hassan et al., 2013; Harun et al., 2015). In contrast, the 7-OH-mitragynine content comprises up to 1.6 percent of the total alkaloid content of the plant (or approximately 0.1-0.3 mg/g in *M. speciosa* leaves) (Kruegel et al., 2016; Warner et al., 2016; Ponglux et al., 1994).

The amount of naturally occurring variations of mitragynine and 7-OH-mitragynine concentrations in *M. speciosa* leaves is unknown because of the limited number of evaluations. For example, concentrations of mitragynine and 7-OH-mitragynine were high in one raw leaf sample (23.8 mg/g and 0.124 mg/g, respectively), while concentrations were low in two other raw leaf samples (0-1.6 mg/g and 0-0.031 mg/g, respectively) (Kikura-Hanajiri et al., 2009). Broad variations of the two alkaloids were also detected in 11 of the 13 finished products (powder and resin) ranging from 1.2 to 6.3 percent for mitragynine and from 0.01 to 0.04 percent for those of 7-OH-mitragynine. The concentrations of the target compounds in the resin samples were much higher than those in the dried leaves (Ruri et al., 2009). The alkaloidal content of *M. speciosa* appears to depend greatly on factors such as growing environment, plant variety, age of

µg/ml, with an accuracy of 97-102 percent. It can quantify mitragynine in dried botanical samples and liquid extracts.

Pharmacokinetics

Distribution

A pharmacokinetic study with mitragynine was conducted in 10 male users of *M. speciosa* who were administered a tea made from 40 grams of fresh leaves boiled for 1 hour in 2 liters of distilled water (Trakulsrichai et al., 2015). Three mitragynine concentrations of tea were prepared: 0.1042, 0.166, and 0.1917 mg/ml. Subjects ingested 60 or 120 ml of the tea every day for 7 days. Notably, all subjects reported experiencing tongue numbness upon drinking the tea, which they said was similar to their use of tea made from *M. speciosa* outside the laboratory.

Pharmacokinetic measurements were taken on the 8th day following one additional dose of the tea. The doses for the first 7 days plus the final 8th day loading dose were: 6.25 mg/day + 12.5 mg (n = 3), 6.25 mg/day + 6.25 mg (n = 2), 9.96 mg/day + 19.92 mg (n = 1), 9.96 mg/day + 9.96 mg (n = 3), 11.5 mg/day + 23 mg (n = 1).

The Cmax for mitragynine ranged from 0.0185 µg/ml for the 9.96 mg loading dose and 0.105 µg/ml for the 23 mg loading dose. The AUC0-tlast ranged from 0.062 µg hr/ml for the 9.96 mg loading dose to 0.67 µg hr/ml for the 23 mg loading dose. Both parameters showed linear pharmacokinetics. Tmax for mitragynine was determined to be 0.83 ± 0.35 hours, with a half-life ($t_{1/2}$) of 23 ± 16 hours. The authors caution that these data may not be representative of individuals who do not have chronic experience with *M. speciosa*. No other pharmacokinetic studies were conducted in humans that evaluate mitragynine or 7-OH-mitragynine. However, Ramanathan et al. (2015) caution that mitragynine may have pharmacokinetic variability based on its hydrophobicity, poor water solubility, high variability of drug release in simulated biological fluids, and acid-degradable characteristics.

Metabolism

In humans, mitragynine produces 7 phase I metabolites through hydrolysis of the methylester in position 16, O-demethylation of the 9-methoxy group and of the 17-methoxy group (Philipp et al., 2009). This is then followed by oxidation to carboxylic acids or reduction to alcohols. Three metabolites were additionally conjugated to glucuronides.

Currently Accepted Medical Use

FDA approves medical use of a drug following the submission and review of a new drug application (NDA) or biologics license application (BLA). FDA has not approved any drug product containing mitragynine or 7-OH-mitragynine for marketing, and neither of these compounds have been studied in humans under an investigational new drug (IND) application.

However, FDA approval of an NDA or BLA is not the only means through which a drug can have a currently accepted medical use in treatment in the United States. In general, a drug may

iii. There must be adequate and well-controlled studies proving efficacy.

"There must be adequate, well-controlled, well-designed, well-conducted, and well-documented studies, including clinical investigations, by experts qualified by scientific training and experience to evaluate the safety and effectiveness of drugs, on the basis of which it could be fairly and responsibly concluded by such experts that the substance will have the intended effect in treating a specific, recognized disorder."

A search of the medical and scientific literature shows that there are no published papers reporting on standardized clinical studies in which purified mitragynine, purified 7-OH-mitragynine, or *M. speciosa* plant-based compounds were administered to evaluate their safety or efficacy. The criteria for an adequate and well-controlled study for purposes of determining the safety and efficacy of a human drug are defined in 21 CFR 314.126.

Therefore, there are not adequate, well-controlled studies showing that the substances will have the intended effect in treating a specific, recognized disorder.

iv. The drug must be accepted by qualified experts.

"The drug has a New Drug Application (NDA) approved by the Food and Drug Administration, pursuant to the Food, Drug and Cosmetic Act, 21 U.S.C. 355. Or, a consensus of the national community of experts, qualified by scientific training and experience to evaluate the safety and effectiveness of drugs, accepts the safety and effectiveness of the substance for use in treating a specific, recognized disorder. A material conflict of opinion among experts precludes a finding of consensus."

This element requires that the national community of experts, qualified by scientific training and experience to evaluate the safety and effectiveness of drugs, accepts the safety and effectiveness of the substance for use in treating a specific, recognized disorder. Medical practitioners who are not experts in evaluating drugs are not qualified to determine whether a drug is generally recognized as safe and effective or meets NDA requirements (57 FR 10499-10505).

There is no evidence that there is a consensus among qualified experts that mitragynine or 7-OH-mitragynine are safe and effective for use in treating a specific, recognized disorder. This is related to the fact there are no adequate scientific studies showing that mitragynine or 7-OH-mitragynine are safe and effective in treating a specific, recognized disorder (as discussed above).

v. The scientific evidence must be widely available.

"In the absence of NDA approval, information concerning the chemistry, pharmacology, toxicology, and effectiveness of the substance must be reported, published, or otherwise widely available, in sufficient detail to permit experts, qualified by scientific training and experience to evaluate the safety and effectiveness of drugs, to fairly and responsibly conclude the substance is safe and effective for use in treating a specific, recognized disorder."

these disorders are often treated in Western countries with pharmaceutical mu opioid agonists, and both mitragynine and 7-OH-mitragynine are also mu opioid agonists.

M. speciosa also has been used for abuse purposes to produce euphoria (Suwanlert, 1975). Individuals chew 1-3 fresh leaves for both abuse and medicinal purposes, reportedly chewing 3–10 times a day (Suwanlert, 1975). Amattayakul (1960) calculated that 20 *M. speciosa* leaves contain about 17 mg of mitragynine (0.85 mg/leaf).

As would be expected from a mu opioid agonist that is unscheduled under the CSA and easily accessible, *M. speciosa* is also being used recreationally as an uncontrolled "legal high" because of its euphorigenic properties (Swogger et al., 2015; Warner et al., 2016).

5. The Drug's Scope, Duration, and Significance of Abuse

The fifth factor the Secretary must consider is the scope, duration, and significance of abuse of mitragynine and 7-OH-mitragynine. Evidence from DEA reports and published reports in the scientific literature demonstrate the scope, duration, and significance of abuse of mitragynine and 7-OH-mitragynine.

Availability of Mitragynine and 7-OH-mitragynine

The primary source of mitragynine and 7-OH-mitragynine in the United States is through Internet sales of *M. speciosa* (Prozialeck et al., 2012; Logan et al., 2012; Hillebrand et al., 2010; Babu et al., 2008; Rosenbaum et al., 2012; Schmidt et al., 2011), although it is also sold in brick-and-mortar shops. The most common forms of *M. speciosa* sold by online vendors are powders, capsules, resin extracts, crushed leaves, and tablets (Warner et al., 2016).

Although *M. speciosa* is often marketed as a dietary supplement (Boyer et al., 2007; Boyer et al., 2008; and Sheleg and Collins, 2011), there is currently no evidence that it is being sold through any of the corporate nutritional supplement chain stores in the United States (Prozialeck et al., 2012).

Poison Control Center Reporting

The Center for Disease Control and Prevention (CDC) analyzed data from the National Poison Data System (NPDS) for reports of exposure to *M. speciosa* over the period of January 2010 through December 2015. The NPDS is maintained by the American Association of Poison Control Centers and records data from poison control center calls received across the United States (including 50 states, the District of Columbia, and Puerto Rico (Anwar et al., 2016)). During the 6-year period of 2010 to 2015, there were a total of 660 calls related to kratom exposure. The number of calls each year rose from 26 in 2010 to 263 in 2015, a 10-fold increase. Cases resulting in consultation with a health professional represented 75 percent of the calls (n = 496). Cases that represented non-life threatening complications, with no residual disability, were reported for 42 percent of the calls (n = 275). Potentially life-threatening complications, some with residual disability, were reported for 7 percent of the calls (n = 49).

the FD&C Act (21 U.S.C. 342(f)(1)(B)) because they contain a new dietary ingredient for which there is inadequate information to provide reasonable assurance that such ingredient does not present a significant or unreasonable risk of illness or injury.

Kratom is a botanical that qualifies as a dietary ingredient under section 201(ff)(1) of the FD&C Act (21 U.S.C. 321(ff)(1)). When marketed as a dietary ingredient, FDA also considers kratom to be a new dietary ingredient under section 413(d) of the Act (21 U.S.C. 350b(d)) because, to the best of the Agency's knowledge, there is no information demonstrating that this substance was marketed as a dietary ingredient in the United States before October 15, 1994. Furthermore, based on FDA's review of the publicly available information regarding kratom, there does not appear to be a history of use or other evidence of safety establishing that kratom will reasonably be expected to be safe as a dietary ingredient. In the absence of a history of use or other evidence of safety establishing that kratom will reasonably be expected to be safe as a dietary ingredient, kratom and kratom-containing dietary supplements and bulk dietary ingredients are adulterated under section 402(f)(1)(B) of the FD&C Act (21 U.S.C. 342(f)(1)(B)) because they contain a new dietary ingredient for which there is inadequate information to provide reasonable assurance that such ingredient does not present a significant or unreasonable risk of illness or injury.

As explained in Import Alert 54-15, shipments of kratom and kratom-containing dietary supplements and bulk dietary ingredients may be refused admission pursuant to 21 U.S.C. 381(a)(3) and section 801(a)(3) of the FD&C Act in that they appear to be a dietary supplement or contain a dietary ingredient that is a new dietary ingredient for which there is inadequate information to provide reasonable assurance that such ingredient does not present a significant or unreasonable risk of illness or injury (section 402(f)(1)(B)). These shipments have come in a variety of forms, including capsules, whole leaves, processed leaves, leaf resins, leaf extracts, powdered leaves, and bulk liquids made of leaf extracts. Importers' websites have sometimes contained information about how their products are used.

Districts were also alerted to the possibility that the labeling, marketing, and promotion of kratom and kratom-containing products may indicate that such products appear to be unapproved and misbranded drugs. For such products, Districts were advised that Import Alert 66-41 ("Detention Without Physical Examination of Unapproved New Drugs Promoted in the U.S.") may be applicable.

FDA Enforcement Actions

FDA has requested the following seizures of kratom products in the United States:

- In August 2016, at the request of FDA, the U.S. Marshals Service seized more than 100 cases of products labeled as containing kratom. The seized kratom products were marketed as unapproved new drugs and misbranded drugs in violation of the FD&C Act.

- In September 2014, at the request of FDA, the U.S. Marshals Service seized more than 25,000 pounds of raw kratom material that was marketed as an unapproved new drug and a misbranded drug in violation of the FD&C Act.

Subjective responses to *M. speciosa* leaves are dose-dependent, such that low to moderate doses (1–5 g) produce a stimulating effect, while moderate to high doses (5–15 g) produce mu opioid agonist-like analgesia, euphoria, constipation, and sedative-like effects (Sabetghadam et al., 2013; Jansen and Prast, 1988). While new users of *M. speciosa* may be able to achieve desired effects by chewing just a few leaves, more experienced users apparently have to increase their dosage to 10-30 leaves or more due to tolerance development. Heavy users may have to chew *M. speciosa* between three and ten times per day. Effects begin within 5-10 minutes after *M. speciosa* is consumed and persist for several hours (Hassan et al., 2013). Regular *M. speciosa* use is associated with addictive disorders, as evidenced by craving and compulsive use (Singh, 2014).

Epileptic Seizures Associated with Mitragynine and M. speciosa

The medical literature contains three reports of epileptic seizures following *M. speciosa* use:

- Seizures followed by coma were reported in a 64-year-old man 30 minutes after *M. speciosa* was used in combination with the anticholinergic plant *Datura* for treatment of pain and reducing opioid withdrawal symptoms. Following arrival at the emergency department (ED), the man experienced a second seizure. Urine drug screening was positive for cannabinoids, tricyclic antidepressants, and oxycodone. A urine specimen analyzed by HPLC/MS detected a mitragynine concentration of 167 ± 15 ng/ml (Nelsen et al., 2010).

- Seizures were reported after *M. speciosa* was used to reduce opioid withdrawal symptoms in combination with modafinil (Boyer et al., 2008).

- Seizures and foaming at the mouth were reported in a 32-year-old man after recreational *M. speciosa* use. Toxicology screens were negative for other commonly used recreational drugs (Roche et al., 2009).

Liver Abnormalities with Mitragynine and M. speciosa

There are two case reports in which *M. speciosa* use resulted in liver damage.

- A 58-year-old man with schizoaffective disorder experienced jaundice and liver injury on two separate occasions (separated by a year apart) following regular use of powdered *M. speciosa* for anti-anxiety purposes (Dorman et al., 2015). In both instances, the man continued taking his psychotropic medication, which was identified as quetiapine (100 mg daily) in the first instance and sertraline (50 mg daily) in the second instance. Liver bilirubin, ammonia, and enzyme levels were all determined to be abnormal and medication was discontinued temporarily. Upon discontinuation of *M. speciosa* in both instances, liver tests were returning to normal after several days. Evaluation of body fluids for mitragynine and 7-OH-mitragynine was not conducted.

- **Antihistamines** (promethazine, diphenhydramine, and alimemazine; n = 6 of 23 cases; 23%)
- **Cannabinoid** (THC; n = 5 of 23 cases; 22%)
- **Stimulants** (amphetamine and propylhexedrine; n = 4 of 23 cases; 17%)
- **Antiepileptics** (gabapentin, carbamazepine, lamotrigine, and divalproex; n = 4 of 23 cases; 17%)
- **Antipsychotics** (quetiapine and olanzapine; n = 3 of 23 cases; 13%)
- **Muscle relaxants** (carisoprodol and cyclobenzaprine; n = 2 of 23 cases; 9%)
- **Hallucinogen** (dimethoxyamphetamine; n = 2 of 23 cases; 9%)
- **Non-steroidal anti-inflammatory** (phenazone; n = 1 of 23 cases; 4%)
- **Cough suppressant** (dextromethorphan; n = 1 of 23 cases; 4%)

Of the 24 fatal cases associated with mitragynine, 7 were published as case reports in the medical literature. These 5 published reports demonstrate the variety of multiple drugs taken in conjunction with mitragynine, which include:

- Venlafaxine, diphenhydramine, and mirtazapine (all at therapeutic levels); in addition, ethanol (peripheral blood ethanol level of 0.02 %) and mitragynine. Mitragynine was present in all fluids and tissue examined (peripheral blood = 0.23 mg/l, central blood = 0.19 mg/l, liver tissue = 0.43 mg/kg, urine = 0.37 mg/l). The cause of death of the 24-year-old male was ruled to be "mixed drug intoxication—primarily mitragynine" (McIntyre et al., 2015).

- Propylhexedrine (a stimulant in decongestant inhalers that was previously in Schedule V but was decontrolled in 1991), morphine, promethazine, acetaminophen, and mitragynine. Mitragynine was present in plasma (0.39 mg/l). Cause of death was ruled to be the result of propylhexedrine toxicity rather than mitragynine exposure because there were, at that time, no published toxic levels of mitragynine (Holler et al., 2011).

- Benzodiazepines and over-the-counter cold medications as well as mitragynine. Mitragynine was present in peripheral blood (0.60 mg/l). The 17-year-old male decedent had a history of heroin abuse and was reportedly using *M. speciosa* to treat chronic back pain. Cause of death was ruled to be "possible kratom toxicity" (Neerman et al., 2013).

- Zopiclone, citalopram, and lamotrigine (all at therapeutic levels) in addition to high concentrations of mitragynine (1.06 mg/l) and 7-hydroxymitragynine (0.15 mg/l). Cause of death was ruled to be the result of *M. speciosa* exposure to mitragynine and 7-OH-mitragynine (Karinen et al., 2014).

- Loperamide (at high doses) and marijuana in conjunction with mitragynine. Mitragynine was present in plasma (0.6 mg/l) and in liver tissue (0.68 and 3.5 mg/kg). Cause of death in two married individuals (a 33-year-old and a 37-year-old of unreported gender) was judged to be primarily due to loperamide (Bishop-Freeman et al., 2016).

muscular pain, itching, cramps, joint pain, diarrhea, restlessness, tension, hostility, aggression, sadness, nervousness, delusion, hallucination, anxiety, depression, moodiness, annoyance, sleeplessness, and intense craving (Suwanlert, 1975; Saingam et al., 2013; Saingam et al., 2014; Saingam et al., 2016; Suwanlert, 1975; Vicknasingam et al., 2010; Hassan et al., 2013; Assanangkornchai et al., 2007a; Ahmad and Aziz, 2012; Singh et al., 2014). In many cases, long-term users had difficulty in giving up regular use (Suwanlert, 1975; Singh et al., 2014).

Anecdotal Case Reports of Physical Dependence of Mitragynine and 7-OH-Mitragynine

The following case reports were published in the medical literature regarding individuals who developed physical dependence after chronic use of *M. speciosa*. These individuals include:

- A 37-year-old woman with a history of chronic use of *M. speciosa* experienced severe opioid-like withdrawal syndrome when she discontinued use. Her symptoms were managed with clonidine and hydroxyzine (Galbis-Reig, 2016).

- A man with alcohol dependence and a history of chronic use of *M. speciosa* reported opioid-like withdrawal symptoms when he discontinued use. His symptoms included anxiety, restlessness, tremor, sweating, craving, itch, and feelings of dread. The symptoms were reported as being "short and benign" (McWhirter and Morris, 2010).

- A patient with a history of chronic use of *M. speciosa* (in addition to hypothyroidism) who reported opioid-like withdrawal symptoms when use was discontinued. These symptoms, which resolved within 3 days, included cramping, abdominal pain, sweating, diarrhea, confusion, and delusion (Sheleg and Collins, 2011).

- A 44-year-old man who used *M. speciosa* for 3 years was seen in the ED seeking substance abuse treatment. At that time, he was consuming approximately 40 grams of *M. speciosa* divided into 4 doses over 24 hours. He reported withdrawal symptoms of cravings, anxiety, restlessness, sweating, and itching. Oral fluids were negative for amphetamines, benzodiazepines, cocaine metabolites, methadone, and opiates (McWhirter and Morris, 2010).

8. WHETHER THE SUBSTANCE IS AN IMMEDIATE PRECURSOR OF A SUBSTANCE ALREADY CONTROLLED UNDER THIS ARTICLE

The eighth factor the Secretary must consider is whether mitragynine and 7-OH-mitragynine are immediate precursors of a controlled substance. Mitragynine and 7-OH-mitragynine are not immediate precursors, as defined by 21 U.S.C. 802(23), of any substance already controlled in the CSA.

Alternately, a Schedule II drug can be considered to have a "currently accepted medical use with severe restrictions" (21 U.S.C. 812(b)(2)(B)). Yet as stated above, the lack of accepted medical use for a specific, recognized disorder precludes the use of mitragynine or 7-OH-mitragynine even under conditions where its use is severely restricted.

(C) There is a lack of accepted safety for use of the drug or other substance under medical supervision.

Since mitragynine and 7-OH-mitragynine have no approved medical use and have not been investigated as new drugs, their safety for use under medical supervision has not been determined. Thus, there is a lack of accepted safety for use of mitragynine and 7-OH-mitragynine under medical supervision.

FDA therefore recommends that mitragynine and 7-OH-mitragynine be controlled in Schedule I of the CSA. NIDA concurs with this recommendation.

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